Víctor Manuel Pineda Núñez, M.Eng.

Geotechnical Engineer | PhD Candidate, UNAM

Pile Foundation Design · Numerical Modeling · Soil Monitoring & Instrumentation

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Profile

Geotechnical engineer with solid grounding in soil mechanics, numerical modeling, and data analysis, applied to both research and large-scale infrastructure projects in Mexico. Experience spans pile foundation design, structural-geotechnical forensic monitoring, vacuum consolidation of soft soils, and stochastic modeling of spatial soil variability. Currently completing a PhD at UNAM (expected 2027) on the design of pile group foundations accounting for spatial variability of subsoil properties in the Valley of Mexico, with the aim of translating these research findings into practical recommendations for current geotechnical design regulations in Mexico City.

Research Interests

Incorporating spatial variability of soil properties into geotechnical design under uncertainty · Modeling pile foundation behavior using the Discrete Element Method (DEM) · Negative skin friction in pile groups · Vacuum consolidation of soft clays · Finite element modeling (Plaxis)

Applied research focus: developing methods and tools that translate research findings into practical improvements for the geotechnical industry, including recommendations for current design regulations.

Featured Publication — ICSMFE 2026

Effects of Spatial Variability of Soft Soil Properties on the Estimation of Negative Skin Friction in Pile Groups
Pineda-Núñez, V.M., Auvinet, G., Rodríguez Rebolledo, J.F. — 21st International Conference on Soil Mechanics and Geotechnical Engineering, Vienna, 2026.

Professional Experience

Instituto de Ingeniería, UNAM · Mexico City · 2016–2019, 2025

Geotechnical and Structural Engineering Coordination. Forensic monitoring of foundation and superstructure behavior for federal government buildings (settlement mapping, tilt analysis, numerical modeling); field evaluation of vacuum-assisted consolidation techniques (drain-to-drain and membrane) at NAIM test embankments, with results published in technical reports and peer-reviewed papers. Ongoing PhD research on pile group foundations under spatially variable soil conditions.

Geotecnia y Supervisión Técnica · Xalapa, Veracruz · Oct–Dec 2019

Geotechnical design for Tren Maya, Phase 1 (Palenque to Tulum): stations, track alignment, bridges, and access structures, based on site investigation data across four sections. Contributed to the project report by quantifying the costs associated with geotechnical works along five sections of the route.

ICA · Mexico City · Apr 2011–Mar 2012

Línea 12 Metro Project. Structural-geotechnical analysis and design documentation for underground stations.

Academic Contributions

Contributing author for textbook materials on Stochastic Processes, Probability and Statistics, and Soil Mechanics, supporting teaching and reference resources for geotechnical engineering education.

Updated: 13 de junio de 2026

Education

PhD in Civil Engineering (Geotechnics) (UNAM, 2020–present)
Design of pile group foundations considering spatial variability of subsoil properties, Valley of Mexico.
Advisor: Dr. Gabriel Auvinet Guichard

M.Eng
. in Civil Engineering (Geotechnics) (UNAM, 2013–2016)
Negative skin friction in pile groups in the lacustrine zone of Mexico City.
Advisors: Dr. Gabriel Auvinet Guichard, Dr. Juan Félix Rodríguez Rebolledo

B.Sc. in Civil Engineering (UNAM, 2006–2011)
Behavior of foundations on expansive soils.
Advisor: M.Sc. Agustín Deméneghi Colina

Technical Skills

Plaxis (Advanced) · GeoStudio (Basic) · MATLAB (Intermediate-Advanced) · Python (Intermediate) · AutoCAD (Intermediate-Advanced) · Surfer (Intermediate-Advanced) · LaTeX (Advanced)

Languages: Spanish (Native) · English (Intermediate-Advanced)

Selected Publications

López-Acosta, N.P., Espinosa-Santiago, A.L., **Pineda-Núñez, V.M.**, Ossa, A., Mendoza, M.J., Ovando, E., Botero, E. "Performance of a test embankment on very soft clayey soil improved with drain-to-drain vacuum preloading technology." *Geotextiles and Geomembranes*. https://doi.org/10.1016/j.geotexmem.2019.103459

Pineda-Núñez, V.M., Auvinet, G., Rodríguez Rebolledo, J.F. "Negative skin friction in pile groups in the lacustrine zone of Mexico City." Reunión Nacional de Mecánica de Suelos e Ingeniería Geotécnica, Mérida, 2016.

Full publication list available via ORCID, Google Scholar, and ResearchGate.

Redes de investigación
